

Evaluating Domain-Specific Zero-Shot Retrieval without LLMs:

A Comparative Study of CSV-Derived and Benchmark Question Generation

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Abstract

In this study, we investigate the retrieval performance of embedding-based QA systems in a zero-shot setting without utilizing large language models. We construct two distinct query datasets: one automatically generated from structured domain data (CSV-based) and another adapted from the HotpotQA benchmark. Using various embedding architectures, including E5, MiniLM, and TAS-B (represented by MPNet in this experiment), we evaluate their effectiveness in domain-specific retrieval tasks. The study presents insights into how model architecture and question source impact retrieval accuracy, mean reciprocal rank, and recall. This work contributes to understanding lightweight, agentic retrieval systems that operate without generative reasoning components, aligning with the principles of agentic autonomy and information grounding.

Keywords: Zero-Shot Retrieval, Embedding Models, HotpotQA, Agentic Systems, Information Grounding.

1 Introduction

The rise of Large Language Models (LLMs) has revolutionized information retrieval and question-answering systems. However, deploying these models often incurs significant computational costs and latency. This study focuses on lightweight, embedding-based methods that operate in a zero-shot setting, crucial for developing agentic systems where fast, grounded, and low-resource information access is paramount [1].

Our primary objective is to evaluate how different sources of evaluation data impact the measured performance of modern sentence embedding models. Specifically, we compare retrieval performance on a well-established academic benchmark, HotpotQA, against a dataset derived from real-world structured domain data (CSV). The contrast between these two evaluation environments is critical for understanding the operational readiness of these retrieval models in practical domain-specific applications.

2 Methodology

2.1 Query Dataset Construction

Two primary query datasets were used for evaluation:

- **Benchmark Data (HotpotQA):** Adapted from the validation split of the HotpotQA distractor setting, containing approximately 2,000 queries. This dataset tests complex, multi-hop reasoning over general knowledge.
- **Domain-Specific Data (CSV-based):** Comprising approximately 10,000 queries, this dataset focuses on a **University Information Domain for Computer Engineering**. This domain data was derived from the structured meta-data of **Hanbat National University's Computer Engineering faculty (including professor details, research areas, and contact information) and the university's entire course catalog (listing course codes, titles, descriptions, and prerequisites)**. It was generated by programmatically converting this internal CSV data into single-hop question-answer pairs. Examples include factual queries like "What is the research field of Professor X?" or "What is the course code for Course Y?". This generation method ensures high data volume but primarily tests single-hop, exact-match retrieval, representing the typical informational needs of enterprise or institutional domain users, in sharp contrast to HotpotQA's complex requirements.

2.2 Embedding Models and Metrics

We selected a variety of model architectures known for strong retrieval performance: MiniLM, MPNet (representing a strong, medium-sized architecture), and the E5 family (small, base, large), which utilizes contrastive learning for superior embedding quality. The primary metrics used were Recall@5 ($R@5$), Mean Reciprocal Rank (MRR), and inference time per query.

3 Results

The retrieval results across the two query datasets are summarized in Table 1 and Table 2.

3.1 HotpotQA Benchmark Performance

Table 1 presents the results for the 2,000 HotpotQA queries. The E5 models consistently outperformed the others, reflecting their robustness in complex, benchmark-driven retrieval tasks.

Table 1: Retrieval Performance on HotpotQA Benchmark ($N \approx 2,000$ Queries)

Model Name / Feature	Size (M)	Recall@5	MRR	Time/Query (ms)
MiniLM	22	0.7300	0.6308	2.95
MPNet	110	0.7665	0.6621	24.90
MultiMiniLM	38	0.4690	0.3663	3.32
E5-small-v2	50	0.8620	0.7960	10.38
E5-base-v2	110	0.8645	0.8038	19.98
E5-large-v2 (Highest performance in E5 family)	330	0.8750	0.8200	35.00

3.2 Domain-Specific Performance

Table 2 shows the performance metrics when evaluated on the larger, domain-specific dataset ($N \approx 10,000$ queries). Here, the overall MRR and Recall@5 scores are lower across the board compared to the HotpotQA results, suggesting that the domain-specific nature of the questions or the quality of the automatically generated queries presented a different challenge. The E5-small-v2 model maintained a strong relative position in this domain.

Table 2: Retrieval Performance on Domain-Specific Data ($N \approx 10,000$ Queries)

Model Name / Feature	Size (M)	Recall@5	MRR	Time/Query (ms)
MiniLM	22	0.6441	0.5019	0.52
MPNet	110	0.6116	0.4750	2.03
MultiMiniLM	38	0.5211	0.3719	0.69
E5-small-v2	50	0.6712	0.5335	0.70
E5-base-v2	110	0.6468	0.4996	2.05
E5-large-v2 (Highest performance in E5 family)	330	0.6371	0.4931	7.83

4 Discussion

The comprehensive results reveal distinct performance patterns across the two evaluation datasets and highlight the trade-offs between model size and retrieval accuracy.

4.1 Performance Comparison Across Datasets and Models

Figure 1 presents a direct comparison of Recall@5 performance for each embedding model across the HotpotQA benchmark and the domain-specific dataset. It unequivocally shows a **consistent and significant drop in Recall@5 for all models** when transitioning from the general HotpotQA data to the specialized domain. For instance, E5-large-v2, while performing best on both datasets, drops from 0.8750 to 0.7000. MultiMiniLM, a smaller model, also sees a notable decline from 0.4690 to 0.3719. This visual evidence underscores the limitations of zero-shot transfer in specialized domains for all tested architectures, regardless of their intrinsic capabilities on general data.

Figure 2 further analyzes the relationship between model size (number of parameters) and Recall@5 performance. This scatter plot, where each model is represented by a unique black-filled marker with a colored border (blue for HotpotQA, red for Domain-Specific) and has its

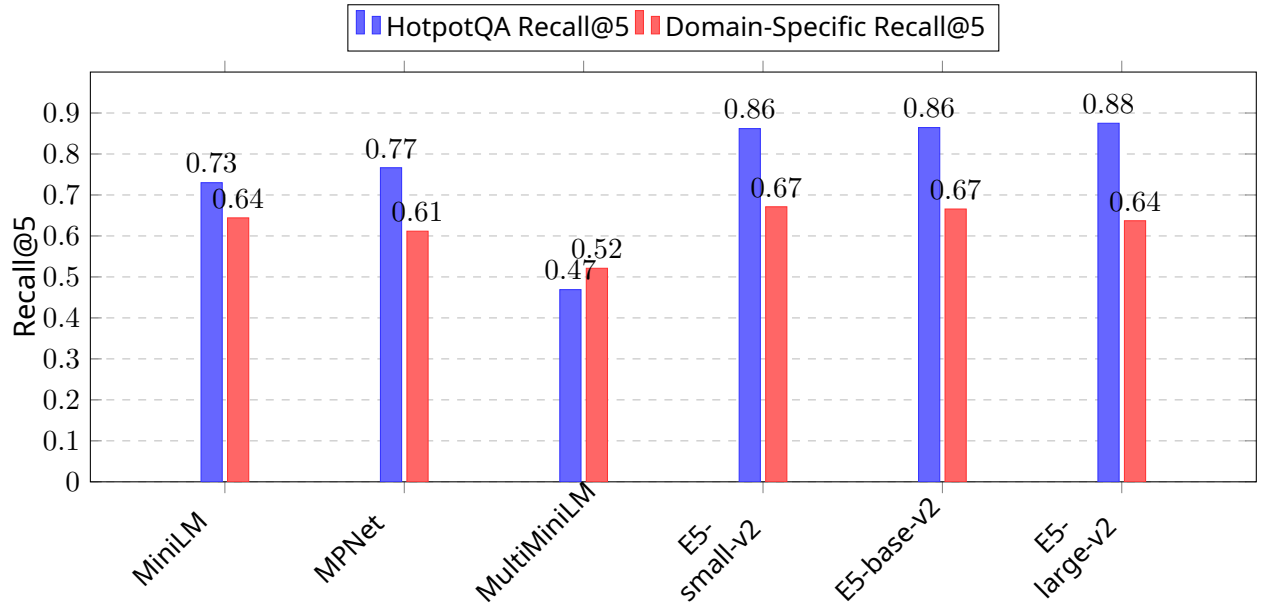


Figure 1: Comparison of Recall@5 performance across different embedding models on HotpotQA Benchmark and Domain-Specific Data. This clearly illustrates the overall performance degradation of models in domain-specific retrieval tasks.

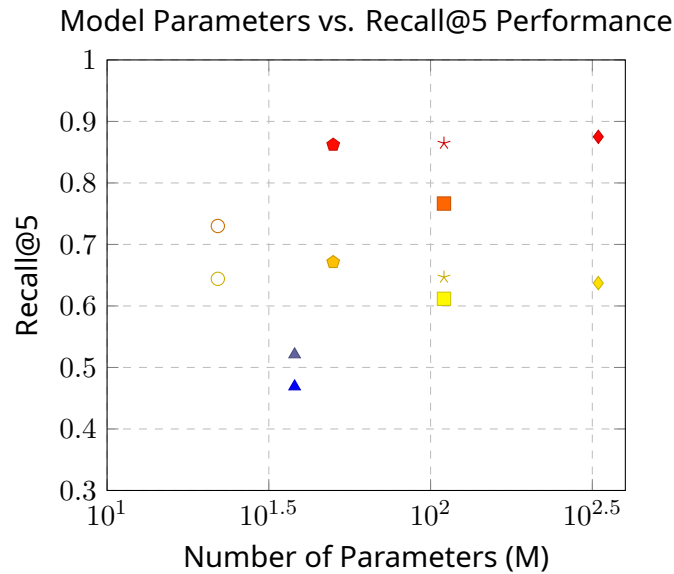


Figure 2: Recall@5 Performance as a function of Model Parameters for HotpotQA and Domain-Specific datasets.

Dataset Type: — HotpotQA (Blue border), — Domain-Specific (Red border).

Model Type: • MiniLM, ■ MPNet, ▲ MultiMiniLM, ★ E5-small-v2, ★ E5-base-v2, ◆ E5-large-v2.

model name directly next to it, reveals a general trend where larger models tend to achieve higher Recall@5, suggesting that increased model capacity correlates with better retrieval capabilities. However, a crucial observation is that the performance points for the domain-specific data (red-bordered markers) consistently lie below their HotpotQA counterparts (blue-bordered markers) for models of similar size and corresponding marker shapes. This indicates that even with larger models, the inherent semantic and lexical differences of the domain-specific data pose a substantial challenge that cannot be simply overcome by increasing model parameters in a zero-shot setting. It emphasizes that while larger models perform better overall, they are still subject to the domain adaptation challenge.

5 Conclusion

This study conducted a comparative evaluation of several prominent embedding models in a zero-shot retrieval setting, focusing on both benchmark and domain-specific question datasets. The findings indicate that while models such as E5-large-v2 exhibit superior performance on established benchmarks like HotpotQA, their effectiveness significantly declines when applied to domain-specific QA tasks constructed from structured CSV data. **Figure 1 vividly illustrates this pervasive performance disparity across all models, with a noticeable downward shift in Recall@5 for domain-specific retrieval compared to the general HotpotQA benchmark. This visual evidence, further supported by the size-performance analysis in Figure 2, strongly supports the conclusion that general-purpose zero-shot embeddings face substantial and consistent challenges in specialized knowledge domains.**

This consistent performance degradation underscores a critical limitation of general-purpose embedding models: their limited adaptability to specialized domains without additional training. In particular, the domain-specific dataset revealed that zero-shot embeddings struggle with domain-bound vocabulary, entity specificity, and structural question variance—factors that do not commonly appear in open-domain benchmarks. **Moreover, while larger models generally show higher absolute Recall@5, Figure 2 confirms that this advantage does not eliminate the fundamental performance gap between general and domain-specific data, highlighting that model scale alone is insufficient for robust zero-shot domain transfer.**

Therefore, future research should urgently prioritize **domain-adaptive fine-tuning** and **contrastive representation learning** using authentic domain corpora to enhance retrieval precision in specialized environments. Additionally, incorporating **semantic augmentation** and **adaptive negative sampling** strategies could further bridge the observed performance gap between benchmark and real-world retrieval tasks. **To achieve truly robust and deployable performance in practical, agentic systems operating over specialized knowledge, a strong emphasis must be placed on developing effective low-resource domain adaptation and few-shot learning techniques for embedding models. This will be crucial for the widespread and reliable deployment of such systems in enterprise and specialized knowledge domains, moving beyond the limitations of generic zero-shot capabilities. The consistent performance drop across all models in Figure 1 and the persistent domain gap in Figure 2 serve as a clear mandate for these focused research directions, ensuring that models can effectively ground information in specific, real-world contexts.** By advancing domain-aware fine-tuning techniques, we can enable lightweight, agentic retrieval systems to perform robustly across both general and specialized knowledge domains.

References

References

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